Pure Mathematics Year 2

Chapter 2

Functions and Graphs

Modulus functions · mappings · domain and range · composite and inverse functions · transformations

1. The modulus function

Solve $|3x - 5| = 2 - \frac{1}{2}x$.

2. Mappings, domain and range

If $f(x) = x^2 - 4x + 3$, find the range of f.

3. Composite functions

For $f(x) = 2x + 1$, $g(x) = x^2$, determine $fg(x)$ and $gf(x)$.

4. Inverse functions

If $f(x) = \dfrac{x + 1}{2x - 1}$, determine $f^{-1}(x)$.

5. Transforming graphs

Sketch $y = |f(x)|$ and $y = f(|x|)$.

Combine transformations and transform the modulus function.

The modulus of a number

The modulus of a, written $|a|$, is its non-negative numerical value.

$|6| = 6, \qquad |-7.1| = 7.1.$

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Distance and magnitude

The distance between two numbers a and b is $|b - a|$, regardless of their order.

For example, the distance between 2 and 7 is

$|7 - 2| = |2 - 7| = 5.$

Modulus measures size: distances and speeds are non-negative. The same notation $|\mathbf{a}|$ gives the magnitude (length) of a vector.

Examples

Luisa · 4

If $f(x) = |2x - 3| + 1$, find:

(a) $f(5)$

a

(b) $f(-2)$

b

(c) $f(1)$

c

Modulus Graphs — Luisa · 5

Sketch $y = |x|$

Complete the values, then sketch the graph.

$x = -2$	$x = -1$	$x = 0$	$x = 1$	$x = 2$

Graph

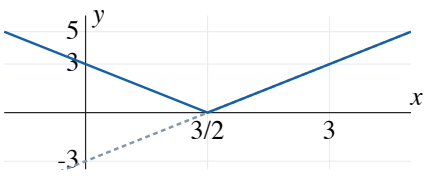
How to sketch $y = |ax + b|$

Sketch $y = ax + b$. Keep the part on or above the x-axis and reflect the part below the axis upwards.

Every output of a modulus function is non-negative.

Modulus Graphs — Luisa · 6

1. Sketch $y = |2x - 3|$



The dotted line is $y = 2x - 3$.
Reflect its negative part upwards.
The vertex is $(\frac{3}{2}, 0)$.

2. Solve $|2x - 3| = 5$

2: Graph

2: Solutions

3. Solve $|3x - 5| = 2 - \frac{1}{2}x$

3: Graph

3: Solutions

4. Solve $|3x - 5| > 2 - \frac{1}{2}x$

4

Test Your Understanding — Luisa · 7

1. Solve $|x+1| = 2x+5$. There is only one solution.

1: Graph

1: Working

2. Solve $|4x-1| < 2x$.

2: Graph

2: Working

Exercise 2A – Y2 Page 26 — Luisa · 8

5. $g(x) = |4 - \frac{3}{2}x|$, $h(x) = 5$.

(a) Sketch both graphs on the same axes.

(b) Hence solve $|4 - \frac{3}{2}x| = 5$.

5a

5b

8. Solve $|3x-5| = 11 - x$.

8: Graph

8

9. (a) On the same axes, sketch $y = |6-x|$ and $y = \frac{1}{2}x - 5$.

(b) State, with a reason, whether $|6-x| = \frac{1}{2}x - 5$ has any solutions.

9a

9b

12. Solve $|2x + 9| < 14 - x$.

12: Graph

12

13. The equation $|6 - x| = \frac{1}{2}x + k$ has exactly one solution.

(a) Find k. (2 marks)

(b) State the solution. (2 marks)

13: Graph

13a

13b

What is a mapping? **Luisa · 10**

What is a mapping?

A mapping relates a set of inputs to a set of outputs. It may follow a rule or be described by individual arrows.

The **domain** is the set of possible inputs. The **range** is the set of outputs that are actually reached.

An arbitrary mapping

$0 \mapsto 4.4$ and 7.2, $2 \mapsto 1$, $3.1 \mapsto 7.2$.

One input can relate to several outputs; different inputs can share an output. Some listed inputs may have no arrow.

Example: $x \mapsto 2x$

x	-1	0	1.7	2
$2x$	-2	0	3.4	4

A general relation may have more than one output for an input, or no output for some listed inputs. A function has a stricter rule, explored next.

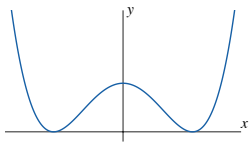
What is a function? **Luisa · 11**

A function maps every element of its domain to exactly one output.

Notation: $f(x) = 2x + 1$ or $f : x \mapsto 2x + 1$. Here $f(x)$ denotes the output for input x.

A graph represents the points $(x, f(x))$. The vertical line test checks whether an input has more than one output.

A. This graph:

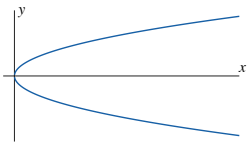


Yes

No

A: Reason

B. This graph:



Yes

No

B: Reason

C. $f(x) = 2^x$

Domain: $\mathbb{R}$.

Yes

No

C: Reason

D. $f(x) = \sqrt{x}$

Domain: $\mathbb{R}$.

Yes

No

D: Reason

E. $f(x) = \pm\sqrt{x}$

Domain: $x \geq 0$.

Yes

No

E: Reason

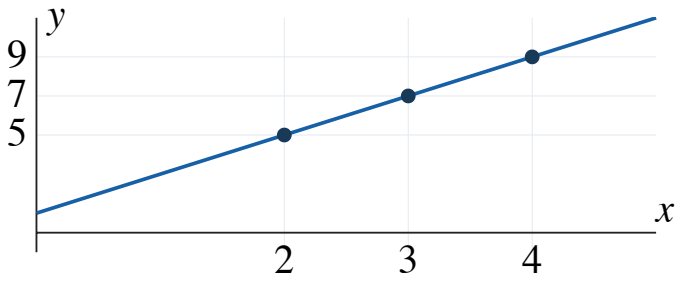
One-to-one vs Many-to-one **Luisa · 12**

Each input of a function has one output. Different inputs may still share the same output. The horizontal line test distinguishes these two types.

One-to-one

Every output in the range comes from exactly one input.

Example: $f(x) = 2x + 1$.

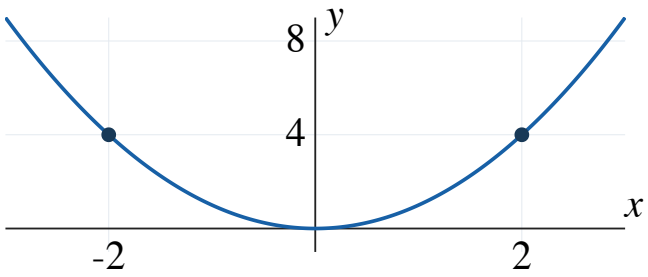


One-to-one

Many-to-one

Two or more inputs can have the same output.

Example: $f(x) = x^2$, $x \in \mathbb{R}$.



Many-to-one

Further Examples

Find each range and state whether the function is one-to-one or many-to-one. Use a sketch where helpful. Write ranges in terms of the output, not the input.

(a) $f(x) = 3x - 2$
Domain: $\{1, 2, 3, 4\}$.

(b) $g(x) = x^2$
Domain: $-5 \le x \le 5$.

(c) $h(x) = 1/x$
Domain: $0 < x \le 3$.

a

b: Sketch

c: Sketch

b: Range

c: Range

Piecewise Functions

A **piecewise function** uses different rules on different parts of its domain.

$$f(x) = \begin{cases} 5 - 2x, & x < 1, \\ x^2 + 3, & x \ge 1. \end{cases}$$

(a) Sketch $y = f(x)$ and state the range.

(b) Solve $f(x) = 19$.

a: Range

a: Sketch

b

Edexcel C4 June 2012, Q6(a)

$f(x) = e^x + 2$, $x \in \mathbb{R}$. State the range of f.

1: Sketch

1: Range

Edexcel C4 June 2010, Q4(d)

$g(x) = x^2 - 4x + 1$, $0 \leq x \leq 5$. Find the range.

2: Sketch

2: Range

7. The function s is defined by

$$s(x) = \begin{cases} x^2 - 6, & x < 0, \\ 10 - x, & x \geq 0. \end{cases}$$

(a) Sketch $y = s(x)$.

(b) Find all a such that $s(a) = 43$.

(c) Solve $s(x) = x$.

7a

7b

7c

8. The function p is defined by

$$p(x) = \begin{cases} e^{-x}, & -5 \le x < 0, \\ x^3 + 4, & 0 \le x \le 4. \end{cases}$$

(a) Sketch $y = p(x)$.

(b) Find all a, to 2 decimal places, such that $p(a) = 50$.

8a

8b

The left branch ends at the open point $(0, 1)$; the right branch starts at the filled point $(0, 4)$.

9. The function h has domain $-10 \le x \le 6$. It is linear from $(-10, 14)$ to $(-4, 2)$, and from $(-4, 2)$ to $(6, 27)$.

(a) Sketch $y = h(x)$.

(b) State the range.

(c) Find all a such that $h(a) = 12$.

9a

9b

9c

Exercise 2B – Y2 Pages 30-32

11. $f(x) = ax^3 + bx - 5$, where a, b are constants. Given $f(1) = -4$ and $f(2) = 9$, find a, b.

11

12. $h(x) = x^2 - 6x + 20$, with domain $x \geq a$. Given that h is one-to-one, find the smallest possible a. (6 marks)

12: Sketch

12

Summary of Domain/Range

Identify the range of each function. Sketch the graph and check whether the domain is restricted.

1. $f(x) = x^2, x \in \mathbb{R}$.

1

3. $f(x) = \ln x, x > 0$.

3

5. $f(x) = x^2 + 2x + 9, x \in \mathbb{R}$.

5

2. $f(x) = 1/x, x \neq 0$.

2

4. $f(x) = e^x, x \in \mathbb{R}$.

4

6. $f(x) = x^2, -1 \leq x \leq 4$.

6

Function or output?

Compare $f : x \mapsto 2x + 1$ and $f(x) = 2x + 1$.

The first describes the mapping f; the second gives its output for input x. A variable can represent a function, just as it can represent a number, vector or set.

Functional equations

Solving $2x + 1 = 5$ finds a number, $x = 2$. Solving

$f(x + y) = f(x)f(y)$

means finding functions that satisfy it.

Example

Functions as inputs

A computer program can accept a function as an input. For example, applying $x \mapsto x + 1$ to every entry of $[1, 2, 3]$ gives $[2, 3, 4]$.

A map operation builds a new list by applying f to each element of the original list.

Composite functions

Applying functions in succession creates a composite function.

$x \xrightarrow{f} f(x) \xrightarrow{g} g(f(x))$

$gf(x) = g(f(x))$: apply f first, then g.

Read a composition from right to left. The order generally matters.

Example

If $f(x) = x + 1$ and $g(x) = 2x$, compare $gf(x)$ and $fg(x)$.

Compare

Examples

Let $f(x) = x^2 + 1$ and $g(x) = 4x - 2$. Replace each input of the outer function with the whole inner expression.

(a) Find $fg(2)$.

a

(b) Find $fg(x)$.

b

(c) Find $gf(x)$.

c

(d) Find $f^2(x)$.

d

(e) Solve $gf(x) = 38$.

e

Further Examples

$f(x) = |2x - 8|$, $g(x) = \dfrac{x+1}{2}$.

(a) Find $fg(3)$.

(b) Solve $fg(x) = x$.

a

b: Sketch

b: Working

Edexcel C4 June 2013(R), Q4

$f(x) = 2|x| + 3$, $g(x) = 3 - 4x$, both on $\mathbb{R}$.

(b) Find $fg(1)$.

(d) Solve $gg(x) + [g(x)]^2 = 0$.

b

d

Edexcel C4 June 2012, Q6(b)

$f(x) = e^x + 2$, $x \in \mathbb{R}$, and $g(x) = \ln x$, $x > 0$.

Find $fg(x)$ in its simplest form.

Answer

3. $f(x) = 3x - 2$, $g(x) = x^2$, $x \in \mathbb{R}$.

(a) Find $fg(x)$.

(b) Solve $fg(x) = gf(x)$.

3a

3b

4. $p(x) = \dfrac{1}{x-2}$, $x \neq 2$, and $q(x) = 3x + 4$, $x \in \mathbb{R}$.

(a) Express $qp(x)$ as $\dfrac{ax+b}{cx+d}$.

(b) Solve $qp(x) = 16$.

4a

4b

5. $f(x) = |9 - 4x|$, $g(x) = \dfrac{3x - 2}{2}$.

(a) Find $fg(6)$.

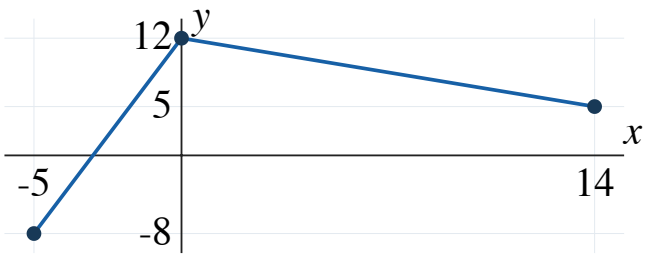
(b) Solve $fg(x) = x$.

5a

5b: Sketch

5b: Working

11. The domain of g is $-5 \leq x \leq 14$. It is linear from $(-5, -8)$ to $(0, 12)$, then to $(14, 5)$.



(a) State the range of g.

(b) Find $gg(0)$.

(c) Given $h(x) = \dfrac{2x - 5}{10 - x}$, find $gh(7)$.

11a

11b

11c

MAT 2014, 1F

$S(x) = x + 1$, $T(x) = -x$. Apply S a total of s times and T a total of t times, in some order, to obtain $8 - x$. What follows?

(i) $s = 8, t = 1$; (ii) s odd, t even; (iii) s even, t odd; (iv) both powers of 2; (v) none.

MAT 2012, Q2

$f(x) = x + 1$, $g(x) = 2x$.

(i) Show $f^2g(x) = gf(x)$.

(ii) Find all compositions giving $4x + 4$.

(iii) Find $f^igf^jgf^k(x)$, for integers $i, j, k \geq 0$.

(iv) How many compositions give $4x + 4m$, where $m \geq 0$ is an integer?

i

Undoing a function

An inverse function reverses the original mapping. If $f(4) = 2$, then $f^{-1}(2) = 4$.

For $f(x) = 2x + 1$:

$x \xrightarrow{\times 2} 2x \xrightarrow{+1} 2x + 1$.

Undo the operations in reverse order: subtract 1, then divide by 2.

$$f^{-1}(x) = \frac{x - 1}{2}.$$

Why must f be one-to-one?

Reason

Notation

f^{-1} means the inverse function; it does not mean $1/f$.

Similarly, $\sin^{-1}$ denotes inverse sine.

Applying f, then f^{-1}, returns the original input.

More on Inverse Functions — Luisa · 31

Method

Write $y = f(x)$, make x the subject, then exchange the variable names.

Example: $y = 2x + 1$ gives $x = \dfrac{y-1}{2}$, hence $f^{-1}(x) = \dfrac{x-1}{2}$.

1. Find the inverse of $f(x) = 3 - 4x$.

1

2. Find the inverse of $f(x) = \dfrac{x+2}{2x-1}$, $x \neq \frac{1}{2}$.

2

Graphing an Inverse Function — Luisa · 32

Reflect in $y = x$

Inverting swaps input and output: (a, b) on $y = f(x)$ becomes (b, a) on $y = f^{-1}(x)$.

The graphs are reflections in $y = x$. Vertical intercepts become horizontal intercepts; vertical asymptotes become horizontal ones.

The domain of f is the range of f^{-1}, and vice versa.

Domain → range

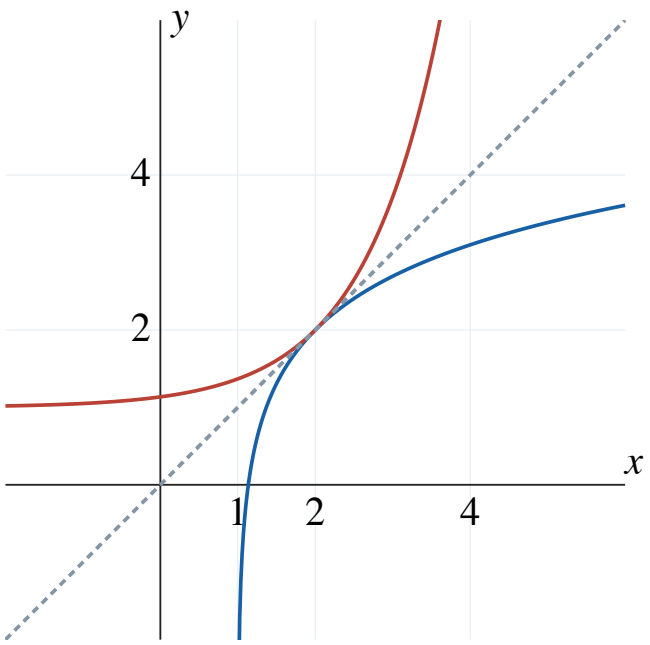


Illustration: blue $f(x) = \ln(x-1) + 2$; red $f^{-1}(x) = e^{x-2} + 1$. The dashed line is $y = x$.

Example

Luisa · 33

$g(x) = \sqrt{x-2}$, with domain $x \geq 2$.

(a) State the range.

(b) Find $g^{-1}(x)$.

(c) Sketch both functions.

(d) State the domain and range of g^{-1}.

a

b

d

c: Sketch

Blue: g. Red: g^{-1}. Dashed: $y = x$.

Further Example

Luisa · 34

$f(x) = x^2 - 3$, with domain $x \geq 0$.

(a) Find f^{-1}.

(b) Sketch both functions and state the domain of f^{-1}.

(c) Solve $f(x) = f^{-1}(x)$.

a / b: Domain

c

b: Sketch

Blue: f. Red: f^{-1}. Dashed: $y = x$.

Edexcel C4 June 2012, Q6

$f(x) = e^x + 2, x \in \mathbb{R}$.

(d) Find f^{-1}, stating its domain.

(e) Sketch both curves on the same axes, giving all axis intercepts.

d

e: Features

e: Sketch

Blue: f. Red: f^{-1}. Dashed: $y = x$.

3. Explain why $g : x \mapsto 4 - x$, with domain $x > 0$, is not identical to its inverse.

Formula

Domains

5. $t(x) = x^2 - 6x + 5$, with domain $x \geq 5$. Find $t^{-1}(x)$.

Answer

Sketch

Blue: t. Red: t^{-1}. The endpoints $(5, 0)$ and $(0, 5)$ are included.

6. $m(x) = x^2 + 4x + 9$, with domain $x > a$, for a constant a.

(a) State the least value of a for which m^{-1} exists.

(b) Find $m^{-1}(x)$.

(c) State its domain.

6b

6a

6c

10. $f(x) = 2x^2 - 3$, with domain $x < 0$.

(a) Find $f^{-1}(x)$, stating its domain.

(b) Find the values of a for which $f(a) = f^{-1}(a)$.

10a

10b

Always check roots after squaring. For a decreasing function, intersections with its inverse need not all lie on $y = x$.

12.

$$f(x) = \frac{3(x+2)}{x^2 + x - 20} - \frac{2}{x-4}, \quad x > 4.$$

(a) Show that $f(x) = \dfrac{1}{x+5}$.

(b) Find the range.

(c) Find $f^{-1}(x)$ and its domain.

12a

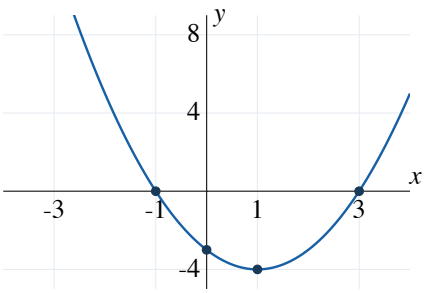
12b

12c

Sketching modulus transformations — Luisa · 41

$f(x) = (x-3)(x+1)$. Sketch $y = |f(x)|$ and $y = f(|x|)$. Show the intercepts and turning points.

Given graph



(a) $y = |f(x)|$

a: Graph

a: Details

(b) $y = f(|x|)$

b: Graph

b: Details

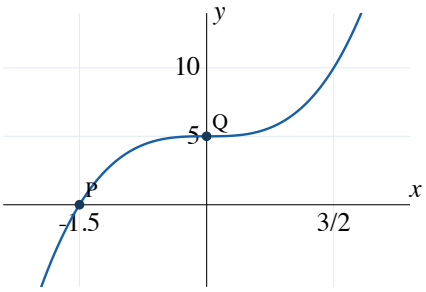
Test Your Understanding — Luisa · 42

Edexcel C4 June 2012, Q4

The given curve passes through $P(-\frac{3}{2}, 0)$ and has a stationary point of inflection at $Q(0, 5)$. Sketch (a) $y = |f(x)|$ and (b) $y = f(|x|)$. Label all axis intercepts. (2 marks each)

The curves below are schematic.

Given graph



(a) $y = |f(x)|$

a: Graph

a: Details

(b) $y = f(|x|)$

b: Graph

b: Details

Further Test Your Understanding — Luisa · 43

Sketch each graph for $-2\pi \leq x \leq 2\pi$. All angles are in radians.

(a) $y = |\sin x|$

a

Reflect every negative part of $y = \sin x$ upwards. All values lie between 0 and 1.

(b) $y = \sin |x|$

b

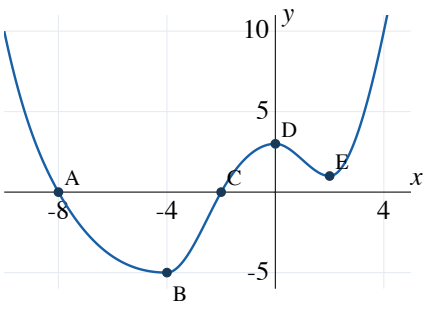
Keep $y = \sin x$ for $x \geq 0$, then reflect this half in the vertical axis. Values are negative when $\pi < |x| < 2\pi$.

Exercise 2E – Y2 Pages 42-44 — Luisa · 44

Exercise 2E, Q5

The schematic graph of f has labelled points $A(-8, 0)$, $B(-4, -5)$, $C(-2, 0)$, $D(0, 3)$, $E(2, 1)$. Sketch the two transformations, labelling corresponding points and axis intercepts. (3 marks each)

Given graph



(a) $y = |f(x)|$

a: Graph

a: Details

(b) $y = f(|x|)$

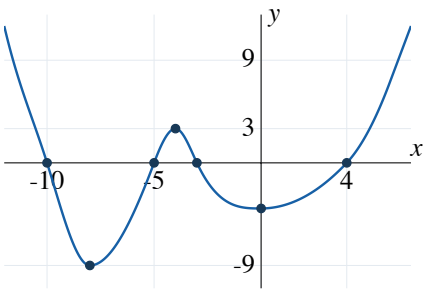
b: Graph

b: Details

Exercise 2E, Q6

The schematic graph of f passes through $A(-10,0)$, $B(-8,-9)$, $C(-5,0)$, $D(-4,3)$, $E(-3,0)$, $F(0,-4)$, $G(4,0)$. Sketch each transformation, labelling the corresponding points and intercepts.

Given graph



(a) $y = |f(x)|$

a: Graph

a: Details

(b) $y = f(|x|)$

b: Graph

b: Details

Exercise 2E, Q10

$$f(x) = \begin{cases} -2x - 6, & -5 \le x < -1, \\ (x+1)^2, & -1 \le x \le 2. \end{cases}$$

(a) Sketch f and state its range. (b) Sketch $|f(x)|$. (c) Sketch $f(|x|)$.

(a) Original function

a: Graph

a: Range

(b) $y = |f(x)|$

b: Graph

b: Details

(c) $y = f(|x|)$

c: Graph

c: Details

SMC 2008, Q25

Find the area of the region satisfying

$\big||x|-2\big|+\big||y|-2\big|\le 4.$

$A:24 \quad B:32 \quad C:64$

$D:96 \quad E:112$

Working

Region

The boundary is included. The same area occurs in all four quadrants.

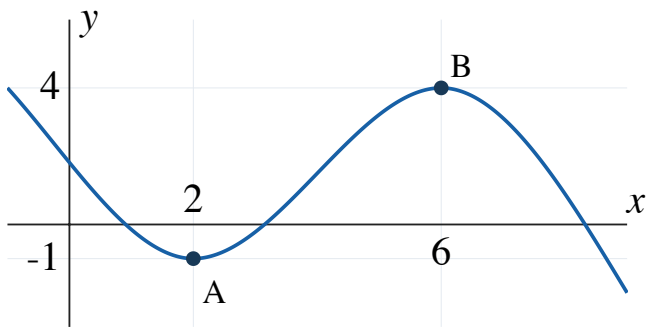
Combining Transformations

Luisa · 48

Combining transformations

A change inside $f(\cdot)$ affects horizontal coordinates; a change outside affects vertical coordinates.

The schematic graph of $y=f(x)$ contains $A(2,-1)$ and $B(6,4)$.



(a) Sketch $y=2f(x+2)$

a: Graph

a: Coordinates

Combining Transformations — Luisa · 49

Continue with the graph containing $A(2, -1)$ and $B(6, 4)$. Sketch each transformed graph. The curves are schematic.

(b) $y = -f(2x)$

b: Graph

b: Coordinates

(c) $y = |f(-x)|$

c: Graph

c: Coordinates

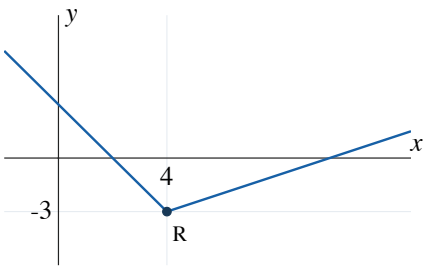
Test Your Understanding — Luisa · 50

Edexcel C4 June 2011, Q3

The graph of $y = f(x)$ consists of two straight branches meeting at $R(4, -3)$. Sketch (a) $y = 2f(x + 4)$, (b) $y = |f(-x)|$. Show the point corresponding to R. (3 marks each)

Diagrams are schematic.

Given graph



(a) $y = 2f(x + 4)$

a: Graph

a: Point

(b) $y = |f(-x)|$

b: Graph

b: Point

What if two x changes or two y changes? — Luisa · 51

Two changes to one coordinate

For $y = 2f(x) + 1$, vertical coordinates become $2y + 1$: multiply by 2, then add 1.

For $y = f(2x + 1)$, solve $2x_{\text{new}} + 1 = x_{\text{old}}$:

$$x_{\text{new}} = \frac{x_{\text{old}} - 1}{2}.$$

Subtract 1, then halve. The order matters!

Sketch $y = \ln(1 - 2x)$

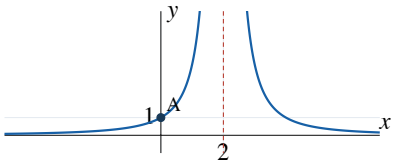
Graph

Details

Exercise 2F – Y2 Pages 47-48 — Luisa · 52

Exercise 2F, Q3

The schematic graph has asymptotes $x = 2$, $y = 0$, and passes through $A(0, 1)$. Sketch each transformation. State the asymptotes and the image of A.



(a) $y = 3f(x) - 1$

a: Graph

a: Details

(b) $y = f(x + 2) + 4$

b: Graph

b: Details

(c) $y = -f(2x)$

c: Graph

c: Details

(d) $y = f(|x|)$

d: Graph

d: Details

Exercise 2F – Y2 Pages 47-48 **Luisa · 53**

Exercise 2F, Q4

$g(x) = (x-2)^2 - 9,\ x \in \mathbb{R}$.

(a) Sketch g, labelling the turning point and intercepts.

(b) Find the image of its turning point under (i) $2g(x-4)$, (ii) $g(2x)$, (iii) $|g(x)|$.

(c) Sketch $g(|x|)$, labelling all turning points and intercepts.

a: Graph

b

c: Graph

a: Points

c: Points

Solving Modulus Problems **Luisa · 54**

Given $f(x) = 3|x-1| - 2,\ x \in \mathbb{R}$:

(a) Sketch the graph. (b) State its range. (c) Solve $f(x) = \frac{1}{2}x + 3$.

Part (a): build the graph in four stages.

$y = |x|$

Start with the basic modulus graph.

$y = |x-1|$

Translate 1 unit right.

$y = 3|x-1|$

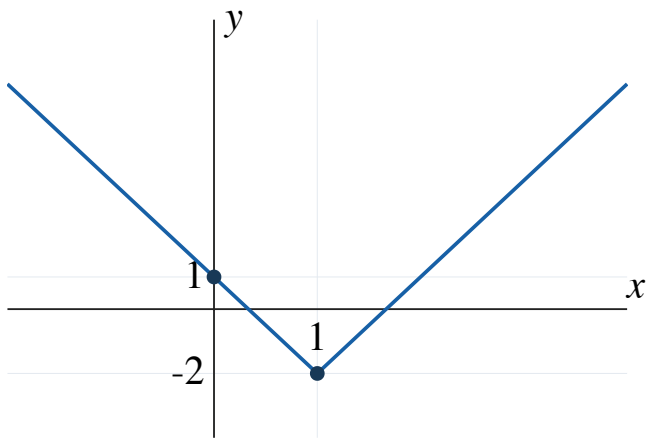
Stretch vertically by factor 3.

$y = 3|x-1| - 2$

Translate 2 units down. Vertex: $(1, -2)$.

$f(x) = 3|x - 1| - 2, x \in \mathbb{R}$.

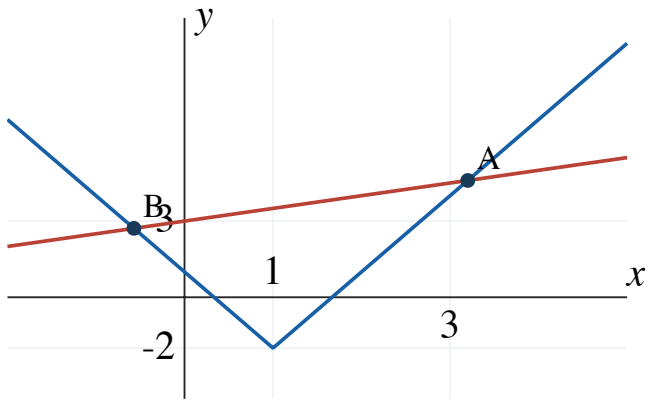
(b) State the range of f.



b

$f(x) = 3|x - 1| - 2$.

(c) Solve $f(x) = \frac{1}{2}x + 3$.



Blue: $f(x)$. Red: $\frac{1}{2}x + 3$. Split into branches at $x = 1$.

c: Right branch

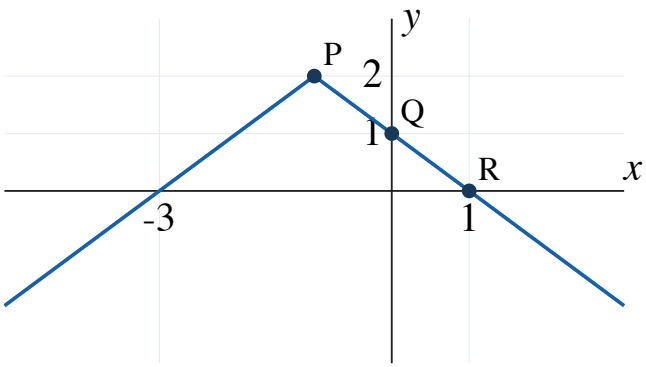
c: Left branch

Edexcel C4 June 2008, Q3

Given $f(x) = 2 - |x + 1|$:

(c) Find P, Q and R. (3 marks)

(d) Solve $f(x) = \frac{1}{2}x$. (5 marks)



c

d

Exercise 2G, Q6

$m(x) = -2x + k, \qquad n(x) = 3|x - 4| + 6.$

Both have domain $\mathbb{R}$. Find the values of k for which $m(x) = n(x)$ has no real roots.

Answer

Example graph

Example shown: $k = 10$. The red line stays below the blue graph.

Exercise 2G, Q7

$s(x) = -10 - x, \qquad t(x) = 2|x + b| - 8.$

Both have domain $\mathbb{R}$. The equation $s(x) = t(x)$ has exactly one real root. Find b.

Answer

Graph

Exercise 2G, Q8

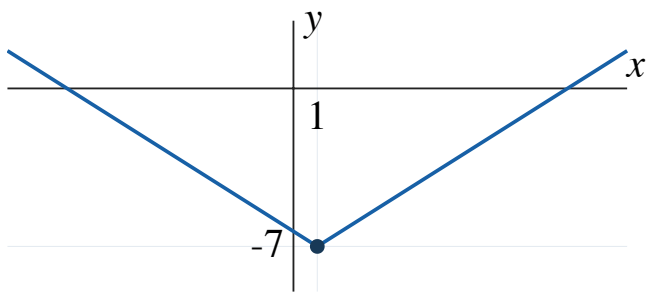
$h(x) = \frac{2}{3}|x - 1| - 7, \quad x \in \mathbb{R}.$

(a) State the range. (1 mark)

(b) Explain why h^{-1} does not exist. (1 mark)

(c) Solve $h(x) < -6$. (4 marks)

(d) Find k for which $h(x) = \frac{2}{3}x + k$ has no solutions. (4 marks)



a

b

c

d

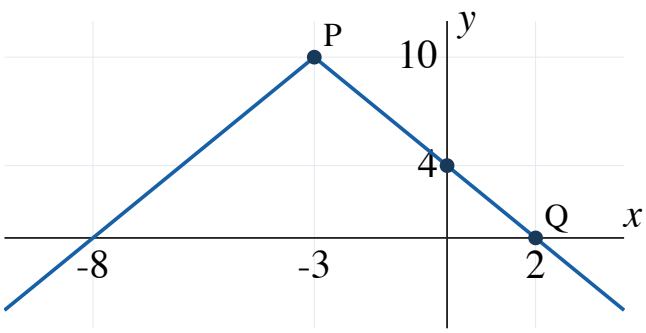
Exercise 2G, Q9

$h(x) = a - 2|x + 3|$, $x \in \mathbb{R}$, crosses the vertical axis at $(0, 4)$.

(a) Find a. (2 marks)

(b) Find the maximum P and right-hand root Q. (3 marks)

(c) Solve $h(x) = \frac{1}{3}x + 6$. (5 marks)



a

b

c

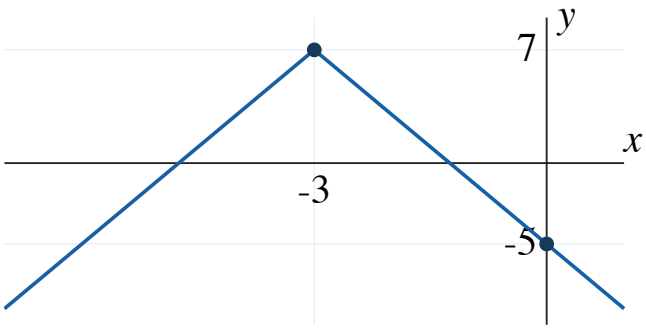
Exercise 2G, Q10

$m(x) = -4|x + 3| + 7, \quad x \in \mathbb{R}$.

(a) State the range. (1 mark)

(b) Solve $m(x) = \frac{3}{5}x + 2$. (4 marks)

(c) Find the values of k for which $m(x) = k$ has two distinct roots. (4 marks)



a

b

c

MAT 2006, Q1I

How many solutions does the equation

$|x| + |x - 1| = 0$

have?

Answer

Graph

Alternatively, $|x| + |x - 1| \geq |x - (x - 1)| = 1$. The graph never reaches the horizontal axis.